01 · THE PROBLEM

Predicting crop yield before planting decisions

Ethiopian crop yield & revenue regression · 5 regions · 5 crops · 2021–2024

The challenge

Predict yield_tons_per_ha for each of the **3,750 leaderboard plots**, using only information known at or before planting time.

A **regression** task scored with **RMSE / MAE / R²** — a number, not a category.

A full workflow, not just a model: integrate → analyse → visualise → evaluate → ship a working demo.

The plot survey alone is not the whole picture: the weather table carries real signal a plot-level model cannot see.

The target

For every leaderboard plot, predict **yield in tons per hectare** — the organizers hold the hidden true yields.

The pipeline in one line

Plot survey
15,090 rows
5 crops · 5 regions

→

+ Weather join
4-month growing season window

→

CatBoost → yield
→ revenue · 108 pipelines, 1 winner

Three raw data sources

Plot survey

15,090 train + 3,750 test plots; region, crop, year, planting month, altitude, fertilizer, improved seed, pest flag, soil quality, labour, distance to market.

Regional weather

Monthly temperature, rainfall and extreme-heat days by region–year–month (~226 rows); joined as a 4-month growing-season window.

Market prices

Price per quintal by crop / region / year; used to turn yield into revenue in the analysis and demo — never a model feature.

Rule 5 — the model uses weather-derived features | **Rule 6** — every median, cap, encoder and scaler is fitted on train only | **Price** is an analysis/demo input, never a feature

Three messy exports → one trustworthy modeling table

Deliverable A — cleaning log, standardised labels, join map, join audit, integrity checks

Data-quality problems found & fixed (A1–A2)

Region labels needed standardising on 93.9% of plot rows — 4+ casing and abbreviation variants (AMH, SOM, …).

Crop spelling: “Tef” → teff; the weather file uses a *different* abbreviation set again.

Two missing-value sentinels (-999 and -999.0) plus true blanks → train-fitted median / mode imputation.

Fertilizer outliers up to ~899 kg/ha → capped at the train 99th percentile (136).

Weather duplicate / conflicting readings averaged; **price unit mix-up** (birr/kg) corrected ×100.

Join logic (A3)

PLOT → seasonal weather (many-to-one; the aggregated weather is one row per plot, so the row count must not change).

PLOT → price (many-to-one, revenue only — never a training feature).

Growing season = planting month + the following 3 months, with Dec→Jan wraparound.

The **plot table is the left table**: every plot must survive the join; weather is only ever aggregated onto it.

A4 join audit — match rates and integrity

15,090 → 15,090
train rows before → after join
(unchanged — as required)

80.6% full
4-month weather window
19.4% partial · 0% unmatched

100%
price match rate
plot → price, both splits

13 / 13
integrity checks PASS
unique ids, no NaNs, ranges

Test set goes through the **same pipeline**, using statistics learned from train only — that is how Rule 6 is enforced structurally, not by promise.

Yield is not value — and heat, not plot inputs, sets the level

Two findings from the analysis (B) and figure pack (C)

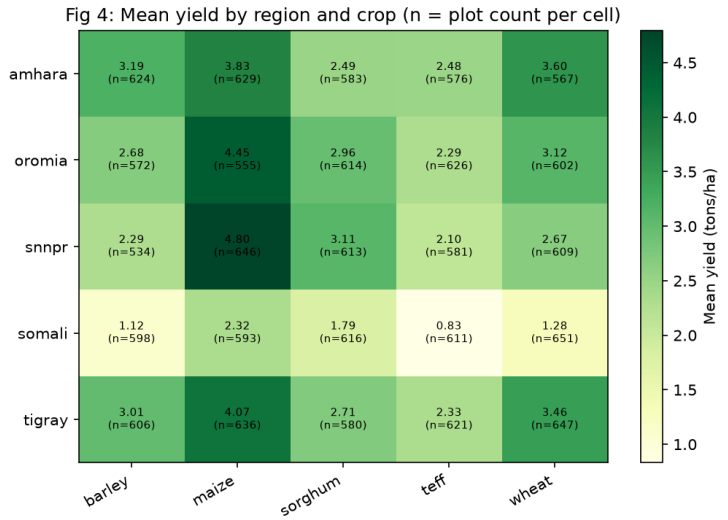


fig04 · Mean yield by region × crop (t/ha)

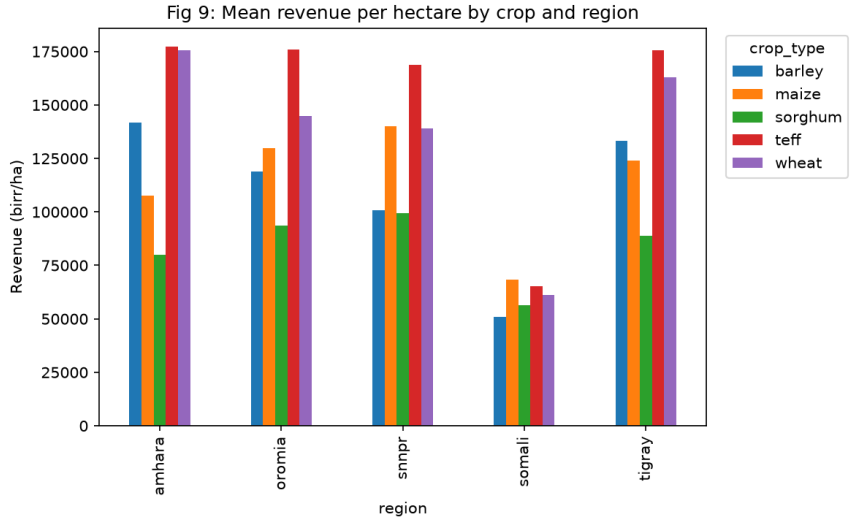


fig09 · Revenue per hectare by crop & region (birr/ha)

FINDING 1 · Yield ≠ value

Teff has the **lowest mean yield (2.00 t/ha)** yet the **highest revenue per hectare (152,301 birr/ha)** — about 1.8× sorghum's 83,623. Maize leads yield (3.90 t/ha) but not revenue. Revenue = yield × 10 quintals/ton × price.

FINDING 2 · Heat sets the level

Somali averages 1.46 t/ha versus ~3.0–3.1 elsewhere. Season temperature and extreme-heat days are the strongest correlates of yield (both ≈ –0.45). Best cell: maize in SNNPR (4.80); worst: teff in Somali (0.83).

108 combinations, one clear winner — and the weather join earns its keep

Deliverable D — baselines, model comparison, cross-validation, out-of-time check, weather ablation

Sweep: **18 models × 2 encoders × 3 scalers = 108 pipelines**, screened on a stratified subsample and confirmed on the full training split. **Winner: CatBoost on log(yield)** (one-hot + robust scaler).

Variant	RMSE	R^2
Mean-predictor baseline	1.413	–0.00
Linear regression	0.808	0.673
CatBoost log-target (winner)	**0.422**	**0.911**

Held-out validation split: 80/20 of the train file, random_state=42.

5-fold CV: 0.427 ± 0.009 t/ha — agrees with the single split within 0.005; the small spread means it is consistently good, not a lucky split.

Out-of-time (2021–23 → 2024): 0.438 t/ha (+0.016) — only minor temporal drift.

Weather ablation: gain of 0.0146 t/ha (3.3%) with the weather features — the growing-season join earns its keep (Rule 5).

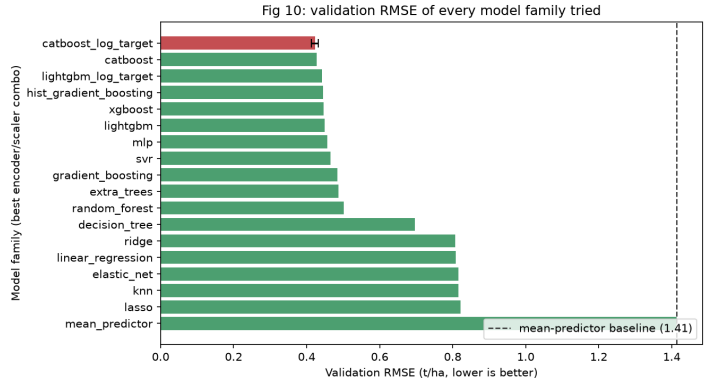


fig10 · Validation RMSE of every model family

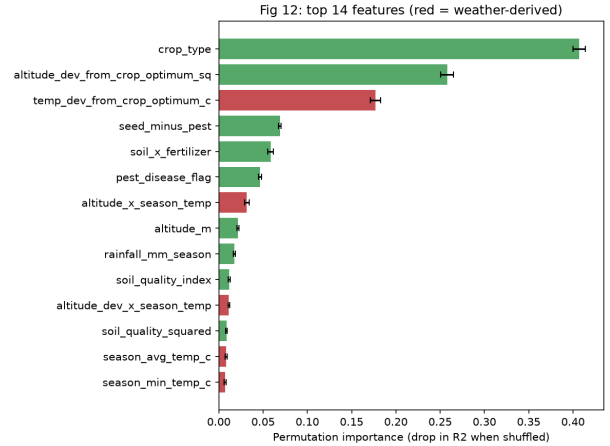


fig12 · Top features (5 of the top 14 are weather-derived)

Where the model still struggles — and the live decision tool

Deliverable D7–D9, the stretch goal, and Deliverable E

RMSE 0.422 t/ha
15.2% of mean yield

MAE 0.304 t/ha
10.9% of mean yield

R² 0.911
variance explained

Error analysis (D7–D8)

Plain language: a typical plot is off by ~0.30 t/ha — about 11% of what it produces, or roughly **14,010 birr/ha** at the average 4,616 birr/quintal price. Good enough to rank plots, not to promise an exact harvest.

By crop: hardest **maize** (0.414), easiest **teff** (0.228). **By region:** Amhara highest (0.355), Somali lowest (0.162) — but relative error is a tight 10.7–11.9%.

Hardest plots cluster at low season temperature and maize: the steep part of the heat-response curve, where a small weather error moves yield a lot.

Stretch — regional equity: after the crop-optimum features the model serves every region within ~1.2 percentage points (10.7–11.9%); a region-calibrated error band is the proposed fix.

Live demo (E) — and what we'd try next

Streamlit app: enter region, crop, year, planting month and plot details — the growing-season weather and the price are **looked up automatically, never typed**. Returns predicted yield, estimated revenue and historical context, with three interactive charts.

Local run: streamlit run app/app.py
streamlit run app/app.py

Next: per-plot prediction intervals (quantile regression / bootstrap), finer sub-regional weather, and more years of data.

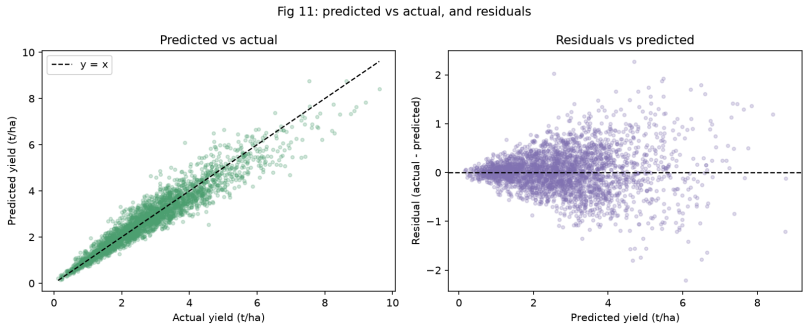


fig11 · Predicted vs actual, and residuals vs predicted (t/ha)